

Jyothi Prasanth

AI ENGINEER | GENERATIVE AI | RAG & MULTI-AGENT SYSTEMS

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PROFESSIONAL SUMMARY

AI Engineer and Data Scientist building production-grade generative AI systems, multi-agent workflows, and retrieval infrastructure. Experienced in LangGraph orchestration, agentic retrieval-augmented generation (RAG), low-latency FastAPI services, automated evaluation, and cloud-native deployment. Delivers measurable gains in latency, faithfulness, model-call efficiency, and operational automation.

TECHNICAL SKILLS

Generative AI & Agents: Claude, Codex, LangGraph, Multi-Agent Orchestration, Agentic RAG, Tool/Function Calling, Model Context Protocol (MCP), Token Context Pruning, Semantic Caching

Retrieval & Evaluation: Qdrant, Pinecone, FAISS (IVF-PQ, HNSW), Hybrid Search, Parent-Document Retrieval, Cross-Encoder Reranking, RAGAS, Promptfoo, LangSmith, LLM-as-a-Judge

AI/ML Engineering: Python, SQL, PySpark, PyTorch, Hugging Face, Sentence Transformers, Scikit-Learn, FastAPI

Cloud & MLOps: AWS (Lambda, SageMaker, DynamoDB, ECS, S3, EFS), Docker, Databricks, MLflow, CI/CD, Feature Drift Monitoring

PROFESSIONAL EXPERIENCE

Flexon Technologies, Inc. - Talent360.ai

Pleasanton, CA

Data Scientist (Generative AI)

Sep 2025 - Present

- Agentic architecture:** Engineered an asynchronous five-node LangGraph workflow for support-ticket triage, with deterministic execution loops and self-correcting reflection nodes that cross-verify claims against retrieved sources.
- Latency and security:** Built FastAPI middleware with automated personally identifiable information (PII) redaction using Regex/NER and Server-Sent Events token streaming to protect sensitive data and reduce time to first token.
- Retrieval optimization:** Implemented hybrid dense-sparse retrieval with metadata pre-filtering on triage signals, reducing average downstream processing latency by 28%.
- Production guardrails:** Instrumented LangSmith tracing and added Promptfoo regression gates to GitHub Actions deployment workflows.

Ziontech Solutions, Inc.

Milpitas, CA

Data Science Intern

Feb 2025 - Aug 2025

- RAG pipelines:** Redesigned ingestion for 10K+ security documents using recursive splitting and parent-document retrieval, increasing RAG faithfulness from 62% to 77% in internal validation.
- Token economics:** Built a FastAPI semantic embedding cache that intercepted 38% of redundant upstream LLM calls; tuned similarity thresholds against offline validation sets.
- Evaluation infrastructure:** Created an automated RAGAS harness tracking context recall, faithfulness, and answer relevance across 500+ stratified test queries.

Iowa State University

Ames, IA

Machine Learning Researcher

Oct 2022 - Dec 2024

- Built a Random Forest pipeline over 50K+ repository records to classify academic content into seven themes, improving accuracy 12% over baseline; added Sentence Transformer embeddings and FAISS validation for another 5% gain.
- Containerized FastAPI classification services on AWS Lambda, automating metadata extraction and tagging while reducing manual curation overhead by 68%.
- Translated model outputs and prediction trends into Tableau dashboards for 11+ enterprise department heads.

KB Technosoft Pvt. Ltd.

Chennai, India

Data Scientist

May 2020 - Dec 2021

- Developed an XGBoost prioritization framework across 80K+ financial records, driving a 9% lift in annual portfolio yield validated through a 90-day A/B test.
- Standardized MLflow experiment tracking and code versioning, and deployed scalable REST APIs to internal systems.

EDUCATION

Iowa State University - MS, Computer Engineering; GPA: 3.78/4.0

Ames, IA | Jan 2022 - Dec 2024

Anna University - BTech, Information Technology; GPA: 3.2/4.0

Chennai, India | Aug 2016 - Mar 2020